

PAPER 1: MEDICINE THROUGH TIME WITH THE WESTERN FRONT

18TH & 19TH C

How much did medicine really change in 18th and 19th Century Britain?

Name: _____

Class: _____

Progress & Assessment Tracker	Effort	Level	Teacher Comments
L1: KT3.1: What breakthrough discoveries changed our understanding of disease causes?			
L2: KT3.2: How did the Industrial Revolution transform prevention and treatment?			
L3: KT3.3: How significant were Edward Jenner and John Snow?			
Final Unit Grade:			

L1: KT3.1: What breakthrough discoveries changed our understanding of disease causes?

LEARNING OBJECTIVES

- ☐ Explain the difference between Spontaneous Generation and Germ Theory.
- ☐ Analyze the significance of Robert Koch's scientific methodology in microbiology.

Do Now Activity

Score: / 7

1. What do historians call the 'Stagnation Paradox' regarding Vesalius and Harvey?

2. What did William Harvey discover about the blood?

3. What did Harvey prove wrong about Galen's theories?

4. In what year was the Great Plague?

5. Explain one way the response to the Great Plague was better than the Black Death.

6. To what extent did Harvey's discovery improve medical treatment in the 1600s?

7. Recall: Who was Andreas Vesalius and what was their key contribution to medicine?

Vocabulary Check

Fill in the blanks using the vocabulary words below.

Germ Theory | Spontaneous Generation | Bacteriology

In nineteenth-century Britain, doctors struggled to agree on the causes of disease. Before Pasteur's breakthrough, the prevailing belief was _____, which claimed that microbes were simply created by decaying matter. However, Pasteur's publication of _____ in 1861 proved that airborne microbes caused decay and illness. This laid the foundation for the new science of _____, led by Robert Koch, who developed scientific methods to isolate and stain specific bacteria under powerful microscopes.

Q1. Identify the scientist who published the Germ Theory in 1861.

Q2. Describe what Spontaneous Generation was.

KNOWLEDGE CHECK (Q3)

How did Pasteur's swan-neck flask experiment disprove the theory of spontaneous generation?

- ☐ A. It showed that liquids only decayed when exposed to airborne microbes, proving that microbes caused decay (not the other way around).
- ☐ B. It proved that microbes were killed by intense cold.
- ☐ C. It showed that diseases were spread by physical contact.
- ☐ D. It proved that microbes spontaneously appeared in a vacuum.

Q4. Explain how Robert Koch developed Pasteur's work.

Q5. Evaluate the significance of the Germ Theory on the history of medicine.

Pair & Share Activity

Prompt: Which was more significant in transforming British medicine: Louis Pasteur's theoretical proof of Germ Theory in 1861, or Robert Koch's practical scientific methods to identify specific disease-causing microbes in the 1870s and 1880s?

Think: Jot down your choice and one piece of evidence from the sources (e.g., Pasteur disproving spontaneous generation and providing the core principles, vs. Koch growing bacteria on agar jelly, staining them with dyes, and finding the TB and cholera microbes).

Your Notes:

You may use the following in your answer:

- You must also use information of your own.**

[illegible]

TEACHER FEEDBACK / D.I.R.T.

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TEACHER FEEDBACK / D.I.R.T.

Exam Q23. 'Louis Pasteur's publication of the Germ Theory was the biggest turning point in understanding the causes of disease.' How far do you agree? Explain your answer. (16 marks)

You may use the following in your answer:

- Spontaneous Generation
- Robert Koch

You must also use information of your own.

General Notes

[illegible]

☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

L2: KT3.2: How did the Industrial Revolution transform prevention and treatment?

LEARNING OBJECTIVES

- ☐ Explain how hospital care and nursing standards were transformed c1700-c1900 under the influence of Florence Nightingale.
- ☐ Analyse how the breakthroughs of anaesthetics and antiseptics solved the critical surgical problems of pain and infection.
- ☐ Evaluate the role of government intervention in disease prevention through the 1875 Public Health Act.

Do Now Activity

Score: / 6

1. What was 'Spontaneous Generation'?

2. Who published the Germ Theory in 1861?

3. What did Robert Koch contribute to Germ Theory?

4. Why did many doctors reject Germ Theory at first?

5. Explain the significance of Koch's methodology (e.g., staining microbes).

6. Recall: Who was William Harvey and what was their key contribution to medicine?

Vocabulary Check

Write a historically accurate sentence connecting two terms from the glossary box below.

Laissez-faire | Aseptic surgery | Compulsory

Your Sentence:

Q1. Identify the first effective anaesthetic discovered in 1847.

KNOWLEDGE CHECK (Q2)

How did Florence Nightingale's 'Notes on Nursing' (1859) help to improve the survival rates of patients in British hospitals?

- ☐ A. It taught nurses how to perform complex internal surgery.
 - ☐ B. It established strict protocols for hygiene, ventilation, and cleanliness, drastically reducing infection rates.
 - ☐ C. It proved that germs caused disease and taught nurses how to use carbolic acid.
 - ☐ D. It taught nurses how to discover new antibiotics.
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Q3. Describe the state of surgery before 1847.

Q4. Explain how Joseph Lister solved the problem of infection.

Q5. To what extent did Simpson's discovery of Chloroform immediately improve surgery?

Pair & Share Activity

Prompt: Was the introduction of anaesthetics (James Simpson's Chloroform) or antiseptics (Joseph Lister's Carbolic Acid) a more significant turning point for patients undergoing surgery in the 19th century?

Think: Jot down your choice and one piece of evidence (e.g., Chloroform solved patient pain but initially led to the 'Black Period' of higher death rates, while Carbolic Acid solved postoperative infection and dropped mortality rates from 46% to 15% but relied on Pasteur's Germ Theory).

Your Notes:

TEACHER FEEDBACK / D.I.R.T.

[illegible]

TEACHER FEEDBACK / D.I.R.T.

Exam Q26. 'The discovery of Carbohic Acid was the most significant turning point in surgery in the years c1700-c1900.' How far do you agree? Explain your answer. (16 marks + 4 marks for SPaG) (16 marks)

You may use the following in your answer:

- Joseph Lister
- Chloroform

You must also use information of your own.

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General Notes

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☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

L3: KT3.3: How significant were Edward Jenner and John Snow?

LEARNING OBJECTIVES

- ☐ Explain the significance and limitations of Edward Jenner's development of the smallpox vaccine.
- ☐ Analyze how John Snow investigated the 1854 cholera epidemic and assess the significance of his findings.

Do Now Activity

Score: / 6

1. Who was Florence Nightingale?

2. What did Nightingale believe caused disease?

3. Name the first effective anaesthetic used in surgery (by James Simpson).

4. Explain the difference between anaesthetics and antiseptics.

5. Why did the 'Black Period of Surgery' occur after the discovery of anaesthetics?

6. Recall: Who was Roger Bacon and what was their key contribution to medicine?

Vocabulary Check

Write a clear definition and a historically accurate sentence for the term: **Inoculation**

Q1. Identify the disease Edward Jenner created a vaccine for.

Q2. Describe how Jenner proved his vaccine worked.

Q3. Explain the actions John Snow took during the 1854 Cholera outbreak.

Q4. Evaluate the significance of the 1875 Public Health Act.

KNOWLEDGE CHECK (Q5)

How did John Snow prove that cholera was a waterborne disease in 1854, long before the discovery of germs?

- ☐ A. By looking at the cholera bacteria under a microscope.
- ☐ B. By mapping the deaths around the Broad Street pump, proving the victims shared a water source rather than breathing the same 'bad air'.
- ☐ C. By boiling all the water in London and watching the disease disappear.
- ☐ D. By injecting a cholera vaccine into local residents.

Pair & Share Activity

Prompt: Whose breakthrough was a more significant turning point in the history of disease prevention: Edward Jenner's 1796 discovery of smallpox vaccination, or John Snow's 1854 discovery that cholera was water-borne?

Think: Jot down your choice and one piece of evidence from the sources (e.g., Jenner's vaccine being the first scientific prevention method that eventually wiped out a disease globally, versus Snow proving the link between physical sanitation and disease, which directly paved the way for Bazalgette's sewers and the 1875 Public Health Act).

Your Notes:

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TEACHER FEEDBACK / D.I.R.T.

Exam Q30. Explain why there was rapid progress in public health in the second half of the nineteenth century (c1850-c1900). (12 marks)

You may use the following in your answer:

- John Snow
- The Great Stink

You must also use information of your own.

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General Notes

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☐ Emerging (1-2) ☐ Emerging+ (3) ☐ Expected (4-5) ☐ Expected+ (6-7) ☐ Greater Depth (8-9)

Teacher Comment:

Factors Overview: 18th & 19th C

Edexcel focuses heavily on the factors that drove medical progress (or held it back). For each factor below, write one specific historical example from this period that either helped or hindered medical progress.

Factor	Specific Historical Example & Impact
Individuals	
The Church & Religion	
Government & Wealth	
Science & Technology	
Attitudes in Society	
War	

Factor	Specific Historical Example & Impact